CS2105 Cheatsheet

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Internet

Switching

- **Circuit Switching** - dedicated physical path (or circuit) is established between two nodes for the duration of a communication session.
- **Packet Switching** - data is broken into small chunks called packets.
  - **Store-and-Forward:** Must receive the entire packet from an inbound link before they can transmit the first bit onto an outbound link.

Network of Networks

Hosts are connected to the Internet via an access ISP. This can provide wired or wireless connectivity. These ISPs are further interconnected, so we have a hierarchy of ISPs. We have **Regional ISPs**, that access ISPs are connected to, **Tier-1 ISPs** which are the highest level of ISP.

Packets

- **Message Segmentation** - breaking up a full message into packets with header containing information.
  - end-to-end delay while using message segmentation is significantly less than sending a big file as one message
  - more tolerant of bit errors, only affected packet needs to be resent
  - but incurs header overhead for each packet
- **Packet Loss** - packets that arrive at a full queue are lost
- **Packet Delay** - made out of processing, queuing, transmission, and propagation delays.
- **Processing Delay** - time taken to examine packet header and check for bit errors.
- **Queuing Delay** - time spent waiting in queue, total bits in queue over link rate/bandwidth. the queuing delay d_{queue} for n packets in queue, packet length L, transmission rate R, and x bits already transmitted:

$$d_{queue} = \frac{nL + (L - x)}{R}$$

- **Transmission Delay** - time taken to push all packets bits onto the wire. $d_{trans} = \frac{L}{R}$ L: packet length (bits)R: transmission rate/bandwidth (bps)
- **Propagation Delay** - time taken for a bit to travel from beginning of the link to the next router. $d_{prop} = \frac{d}{s}$ d: physical distance of the links: propagation speed of the medium ($\approx 2 \times 10^8$ m/s in fiber)
- Total Delay $d_{total} = d_{proc} + d_{queue} + d_{trans} + d_{prop}$

Application Layer

Hosts - run network connections. Either client - initiates connection, or server - waits to be contacted.
Sockets - link between application and transport layer

Web and HTTP

Round Trip Time (RTT) - time for a packet to travel from client to server and go back
Application Architectures:
- **Client-Server** - client initiates communication with the server which is always on
- **Peer to Peer** - hosts act as client and server at the same time
- **Hybrid** - both i guess

Connection Types:
- **Persistent**: leaves connection open after sending response, subsequent HTTP over the same connection. Has 'Connection: Keep-Alive'
  - **Response Time** - the first object takes $2RTT$ (handshake + request). subsequent $N-1$ objects only need $1RTT$ each because the connection is already open.

$$\text{total time} = 2RTT + d_{trans,1} + \sum_{i=2}^{N}(RTT + d_{trans,i})$$

  - **Pipelining** - the client sends multiple http requests back-to-back without waiting for the responses to arrive. $\approx 3RTT + \sum d_{trans}$ for N objects
  - **Multiplexing** - breaking messages into small, independent frames. these frames are interleaved on the same tcp connection, reassembled by receiver on the other end. $3RTT + \max(d_{trans})$ for N objects
- **Non-Persistent**: each object requires a new TCP handshake, adding a delay of one RTT
  - **Response Time** - total time $= N \times (2RTT + d_{trans})$

Stateless - treats every request as an independent transaction. HTTP is stateless, uses cookies to maintain state. **Requests:**
- **GET** - retrieves information from server. Conditional GET - ask if object has been modified, if not, don't send.
- **POST** - send information to server.

Domain Name Service (DNS)

Translates between hostname and IP address. Maintains a resource record that maps between them. Vulnerable to cache poisoning. Runs on UDP, has DNSSEC for security.
DNS Resource Record - Name, Value, Type, TTL
We also have local DNS servers that belong to ISPs, called default name servers, that speed up DNS queries by providing caching. These cache entries can expire after some time based on the TTL.
DNS Name Resolution
1. **Iterative Query**: Local DNS server goes back and forth with each server in the hierarchy.
2. **Recursive Query**: Server that receives the query would be the one doing the next query.

Socket Programming

Sockets are the abstraction interface between processes and transport layer protocols. Consists of:
- IP Address
- Port Number
- Protocol

Stream Socket - for TCP, creates a new welcoming socket that listens for incoming connection requests, and when a client connects, server creates new specific socket for that client so it can listen for others.
Datagram Socket - for UDP, no handshake or flow control, just one socket to send/receive from.
Multiplexing - gathers data from different sockets, wraps with header information, and passes segments to network layer
Demultiplexing - the host receives a segment and uses the header fields to determine which socket should receive the data.
UDP: uses only the destination port number.
TCP: uses a 4-tuple: (source ip, source port, destination ip, destination port).

Reliable Protocols

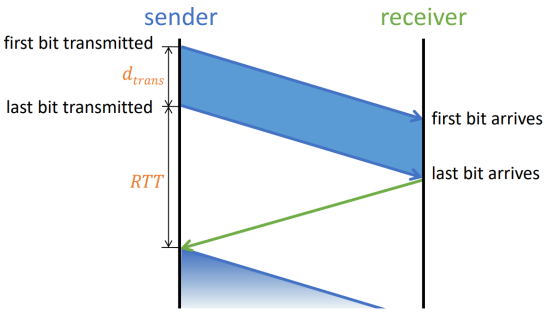
Reliable Delivery Transfer (rdt)

rdt 1.0 - Sender packages data into a packet and sends it, receiver receives the packet, extracts data, and delivers it.
rdt 2.0 - Bit errors are possible. Use checksum to detect bit errors. Using ACK - acknowledgements to tell sender that packet is OK. NAK - negative acknowledgements, tells sender that packet has errors. Sender retransmits. But if NAK or ACK is corrupted, no way to recover.
rdt 2.1 - adds sequence number, receiver then discards duplicate packets.
rdt 2.2 - Replace NAK with ACK of correctly received last packet
rdt 3.0 - handles packet loss or corruption. Wait for a reasonable amount of time for ACK, if no ACK is received, retransmit. Assumes won't reorder packets.

Utilization

The fraction of time the link is actually being used. Calculated by time spent sending / total time.

Problem with Stop-and-wait

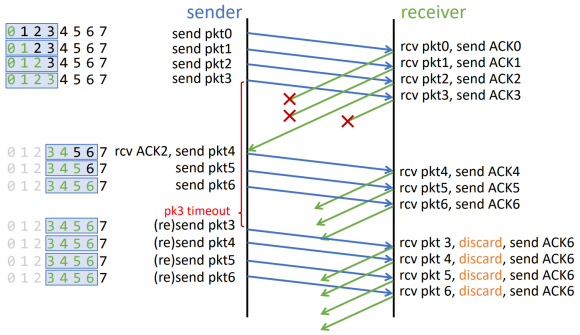


$$U_{sender} = \frac{Time\ sending}{Total\ time} = \frac{d_{trans}}{d_{trans} + RTT}$$

Pipelined Protocols

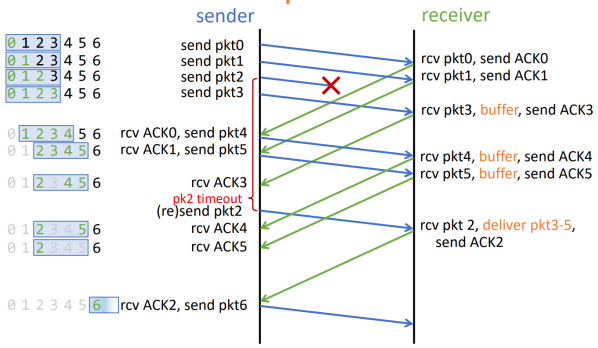
Go-Back-N uses cumulative ACK, where ACK(n) means all packets $\leq$ n have been received. Keep track of ACKed packets, on timeout, retransmit all packets in the window.

Go-back-N in action



Selective Repeat - set a timer per packet, and individually acknowledges all correctly received packets while buffering out of order packets. If time out, resend the packet and restart the timer.

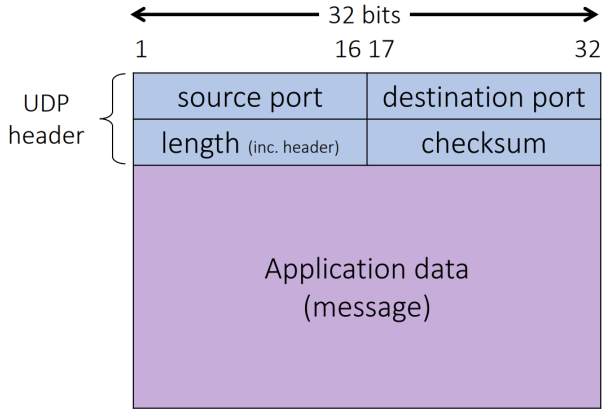
Selective Repeat in action



Transport Services

User Datagram Protocol (UDP)

Unreliable transmission, connectionless. Header consists of 8 bytes:
- Source Port
- Destination Port
- Message Length
- Checksum



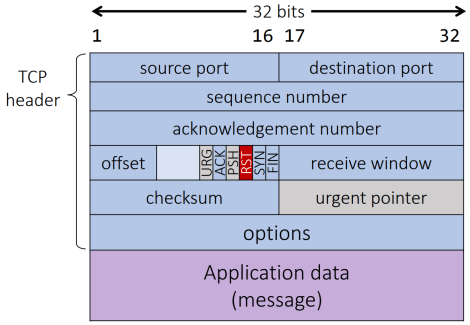
Checksum

8 bit number to detect errors introduced during transmission or storage. Sender and receiver both calculate the checksum to see if they tally.
Calculation:
- Split message into 16 bit integers
- Add integers with a wrap around carry
- Compute 1s complement

Transmission Control Protocol (TCP)

Reliable transmission, connection oriented. Has flow and congestion control.
Header:

TCP segment structure



Consists of:
- **Sequence Number** - "byte number" of the first byte of data in the segment. It allows the receiver to reassemble data in the correct order and detect missing or duplicate packets.
- **Acknowledgement Number** - Sequence number of the next byte of data expected. Cumulative ACK.
- **Offset** - 4-bit field in the TCP header that indicates where the header ends and the actual data (payload) begins.
- **Options** - optional fields appended to the end of the standard 20-byte tcp header.
- **TCP Flags** - SYN, ACK, FIN, PSH, RST, URG
- **Receive Window** - tells sender how much it can send. If field is zero, sender stops.

Maximum Segment Size (MSS) is the largest amount of application-layer data (in bytes) that a communication device can receive in a single TCP segment. Depends on **Maximum Transmission Unit**, and is generally $1500 - 20 - 20 = \mathbf{1460}$ bytes, removing bytes for TCP and IP headers.

Timeout Value

We want timeout to be longer than RTT. We estimate RTT by using an exponential weighted moving average.

$$RTT_E = (1 - \alpha) \cdot RTT_E + \alpha \cdot RTT_S$$

, where $\alpha = 0.125$
Then we compute deviation of RTT

$$RTT_{dev} = (1 - \beta) \cdot RTT_{dev} + \beta \cdot |RTT_S - RTT_E|$$

, where $\beta = 0.25$
To find Retransmission Time Out (RTO) with a safety margin.

$$RTT_E + 4 \times RTT_{dev}$$

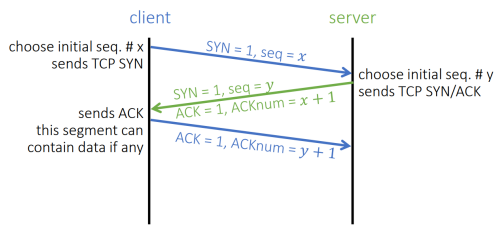
TCP Fast Retransmission (rxmt)
Sender monitors for duplicate ACKs, if receives the same ACK 3 times, assumes the packet after that acknowledgment was lost and immediately retransmits it—without waiting for the timer to expire.

TCP Handshake

Client sends SYN and chooses sequence number x. Server sends back SYN and ACK, with sequence number y and acknowledgement number being $x + 1$. Client then responds with ACK, acknowledges y by sending acknowledgement number as $y + 1$.

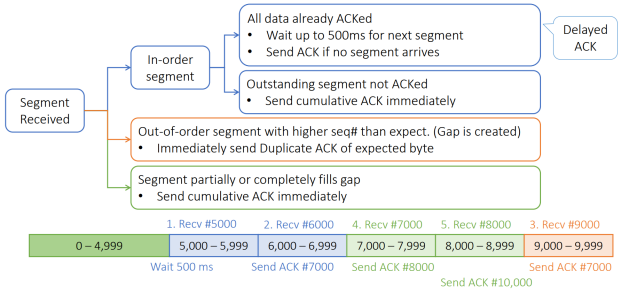
Established using a 3-way handshake

- Agree on connection and exchange parameters



TCP Closing Connection - sends segment with FIN bit set
TCP Flow Control - uses rwnd to manage how much data it can send
TCP Receiver Events

TCP Receiver Events



Network Layer

How do we actually route the packets to the hosts? We need a globally unique address.

IP Address

A globally unique 32 bit binary number that is converted into dotted decimal format. Address format is a.b.c.d/x, x is the number of bits in subnet prefix of IP address.
Associated with **EVERY INTERFACE** of host or router. **Interface** - connection between host and physical link, a router typically has multiple interfaces
Consists of 2 parts:
- **Subnet Prefix** - arbitrary length, n bits.
- **Host ID** - Let $m = 32 - n$, means that there are 2^m addresses in this subnet

Routers contain a forwarding table to forward packets to the correct address. We have **Address Aggregation** to group together IP addresses in the same organisation.

Subnets

A network formed by a group of directly interconnected **host interfaces**. Connect to the outside world through a router. Same subnet can physically reach each other.
Note that host interfaces = not just hosts. So the link between routers is a subnet.
Subnet Mask - Determine which subnet an IP address belongs to. Set all subnet prefix bits to 1, "AND" with the IP.

Network Address Translation (NAT)

Maps multiple private IP addresses within a local network to a single public IP address before transferring information to the internet.
NAT Translation Table - maps a WAN side IP address to a LAN side IP address. Can support up to $2^{16} = 65,536$ simultaneous connections
Implementation
- Replace (source IP address, port number) of every outgoing datagram to (NAT IP address, new port number).
- Remember (in NAT translation table) the mapping from (source IP address, port number) to (NAT IP address, new port number).
- Replace (NAT IP address, new port number) in destination fields of every incoming datagram with corresponding (source IP address, port number) stored in NAT translation table.